

Grade 6 Accelerated
Unit 11
Lesson 7 Homework

Name Answer Key
Date _____
Period _____

Use the scenario for questions 1-3.

Pierson and his friends plan to use a rideshare service for his birthday excursion to International Drive in Orlando, Florida. The service has a booking fee of \$5.75 and charges \$0.46 for the first $\frac{1}{2}$ mile and \$0.26 for each additional 0.2 mile after the first $\frac{1}{2}$ mile.

1. How much will Pierson pay if his trip to International Drive is 23.45 miles from his home?

$$\begin{array}{r} 23.45 \\ -0.5 \\ \hline 22.95 \end{array} \quad \begin{array}{l} \$5.75 + \$0.46 + \$0.26 \left(\frac{22.95}{0.2} \right) \\ \$5.75 + \$0.46 + \$0.26(114.75) \end{array} \quad \begin{array}{l} \$5.75 + \$0.46 + \$29.835 \\ = \$36.05 \text{ total} \end{array}$$

2. Pierson decides to pick up an additional friend. When they arrive at the friend's house, she is not ready to leave. The rideshare service charges \$0.75 for each minute they have to wait. The total bill was \$44.30. How long did Pierson's friend take to get ready?

$$\begin{array}{r} 44.30 \\ -36.05 \\ \hline 8.25 \end{array} \quad \frac{8.25}{0.75} = 11 \quad \text{The friend took 11 minutes to get ready.}$$

3. Pierson's friends decide to use the same rideshare service to take everyone home. They drop the first friend off at his home 19.6 miles from International Drive. The second friend lives another 2.42 miles and Pierson lives another 6.8 miles away. He decides to tip the driver $\frac{2}{9}$ of the bill. How much did Pierson pay for the service, including tip?

$$\begin{array}{r} 19.6 \\ +2.42 \\ +6.8 \\ \hline 28.82 \end{array} \quad \begin{array}{r} 28.82 \\ -0.50 \\ \hline 28.32 \end{array} \quad \begin{array}{l} 5.75 + 0.46 + 0.26 \left(\frac{28.32}{0.2} \right) \\ 5.75 + 0.46 + 36.816 \\ 43.03 \text{ plus tip} \Rightarrow 43.03 \cdot 1\frac{2}{9} = \$52.29 \text{ total with tip} \end{array}$$

4. Ian purchased $1\frac{5}{6}$ pounds of tomatoes, $3\frac{1}{3}$ pounds of onions, 5.3 pounds of avocado and 2 limes to make homemade guacamole.

Produce	Price (per pound)
Tomatoes	\$0.89 per pound
Onions	\$0.65 per pound
Lime	\$0.75 each
Avocado	\$1.09 per pound

Based on the listed prices, how much would Ian spend?

$$1\frac{5}{6} \cdot 0.89 = 1.63 \quad 3\frac{1}{3} \cdot 0.65 = 2.17 \quad 5.3 \cdot 1.09 = 5.78 \quad 2 \cdot 0.75 = 1.50$$

Ian spent \$11.08 total.

Use the scenario for questions 5-7.

Ethan offers a tutoring service for elementary students. He charges an initial fee of \$12.50 for the first tutoring session and provides service for 1.5 hours. After the first session, he charges \$6.75 for $\frac{3}{4}$ hour and charges \$1.86 for each additional $\frac{1}{6}$ hour after the first $\frac{3}{4}$ hour.

5. Ethan tutored a new client this week. The first session was held on Monday, then tutored the client on Wednesday and received payment totaling \$38.78. How many hours did he tutor on Wednesday?

$$\begin{array}{r} 38.78 \\ -12.50 \\ \hline 26.28 \end{array} \quad \begin{array}{r} 26.28 \\ -6.75 \\ \hline 19.53 \end{array} \left(\frac{3}{4} \text{ hour} \right) \quad \frac{19.53}{1.86} = 10.5 \frac{1}{6} \text{ hour} \quad \frac{3}{4} + \left(10 \frac{1}{2} \cdot \frac{1}{6} \right) = 2.5 \text{ hours}$$

6. Ethan tutored a second new client this same week. The first session was held on Tuesday, then tutored the client on Thursday for 1.6 hours. On Thursday, how much would he receive for the total tutoring of this client?

$$\begin{array}{r} 1.6 \\ - .75 \\ \hline 0.85 \text{ hours} \end{array} \quad \frac{0.85}{\frac{1}{6}} = 5.1 \quad \begin{array}{r} 12.50 + 6.75 + 5.1(1.86) \\ 12.50 + 6.75 + 9.486 \\ \hline \$28.74 \end{array}$$

7. If Ethan has a third new client this week and only provides the initial tutoring session, how much money in total has he made for the week for all three clients?

$$38.78 + 28.74 + 12.50 = \$80.02 \quad \text{Ethan made a total of } \$80.02$$

8. A candy store keeps track of the weekly changes in their inventory. Candy is delivered to the shop in $25\frac{1}{8}$ -pound (lb) bags, 16 bags at a time twice a week. The inventory at the end of week shows a total of $133\frac{3}{4}$ lbs. If the same amount of candy was sold each day, how many pounds of candy did they sell in one day?

$$32 \cdot 25\frac{1}{8} = 804 \text{ total bags (lbs. received)} \quad \frac{670.25}{7} = 95.75 \text{ lbs. were sold each day}$$

$$804 - 133\frac{3}{4} = 670.25 \text{ lbs. sold}$$

9. Charlie is building birdhouses for the neighborhood cardinals. To make one birdhouse, she needs 4 pieces of wood $\frac{5}{6}$ feet for the walls, 1 piece of wood 0.33 feet for the floor and 2 pieces of wood $\frac{5}{12}$ feet for the roof. A single board is 9 feet long and costs \$9.32. If Charlie has \$30, how many birdhouses can she make?

$$4 \left(\frac{5}{6} \right) + 0.33 + 2 \left(\frac{5}{12} \right) = \frac{10}{3} + 0.33 + \frac{5}{6} = \frac{1349}{300} \text{ ft./house} \quad \frac{30}{9.32} = 3 \text{ boards she can buy}$$

$$9 \div \frac{1349}{300} = \frac{2700}{1349} = 2 \frac{2}{1349} \text{ birdhouses per board} \quad 3 \cdot 2 \frac{2}{1349} \approx 6 \text{ birdhouses}$$

10. Camila is making bows to sell to support a local nonprofit organization. She purchased a roll of ribbon that is 38.75 ft in length and each bow uses $2\frac{7}{12}$ feet to make. If she makes 11 bows, how many more bows can be made with the ribbon left in the roll?

$$11 \cdot 2\frac{7}{12} = 28\frac{5}{12} \text{ ft used} \quad 10\frac{1}{3} \div 2\frac{7}{12} = 4$$

$$38.75 - 28\frac{5}{12} = 10\frac{1}{3} \text{ ft left} \quad \text{Camila can make 4 more bows.}$$